Extending A358321 by prime-power enumeration and arithmetic exclusions

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Abstract

For $n \geq 3$, let $a(n)$ be the least positive index k for which the n-gonal number $P_n(k) = k((n-2)k-(n-4))/2$ has exactly n distinct prime factors. Subject to the recorded producer coverage and exact arithmetic exclusions described below, we report four locally validated terms of OEIS A358321 for $n = 16, 17, 18, 19$, with $a(16) = 119535945060$, $a(17) = 1113316946280$, $a(18) = 4278672701330$, and $a(19) = 153290693815740$. The search constructs products of increasing distinct prime bases with arbitrary positive exponents, following the constructive method published by Daniel Suteu. Minimality rests on the recorded exhaustive producer coverage; for $n = 19$ it also uses exact exclusions of unfinished subtrees. Indeed, for $p = 3$ or 5, divisibility of $P_{19}(k)$ by p forces divisibility by p^2. Applied after the search, this observation completes the mathematical coverage without asserting native program completion. Independent Python and PARI/GP certificates verify the positive witnesses, not a second exhaustive negative enumeration. A separate final interval $(166539042068893, 170000000000000]$ contains no matching index under the same execution and coverage assumptions. The corresponding four polygonal values give same-index consequences for A358862; neither this interval nor those consequences determine $a(20)$.

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1 Definition, provenance and contribution

For integers $n \geq 3$ and $k \geq 1$, write

$$P_n(k) = \frac{k((n-2)k-(n-4))}{2} = k + (n-2)\frac{k(k-1)}{2}. \tag{1}$$

This is the standard n-gonal number formula [4]. For a positive integer v, let $\omega(v)$ count its distinct prime divisors, with $\omega(1) = 0$; $\Omega(v)$ counts them with multiplicity [5]. The target sequence is

$$a(n) = \min\{k \geq 1 : \omega(P_n(k)) = n\}, \tag{2}$$

whenever the set is nonempty. Thus the parameter n controls both the polygonal order and the required support cardinality. For example, $P_3(11) = 66 = 2 \cdot 3 \cdot 11$ and $a(3) = 11$.

A358321 was submitted by Ilya Gutkovskiy on 12 December 2022 [1]. The source prefix consulted before the computation comprised $n = 3, \ldots, 15$:

$$11, 210, 87, 228, 1155, 7854, 66612, 395646, 2193303, 8389010,$$
$$122574155, 630341910, 6066475415. \quad (3)$$

The official OEIS export retrieved through the OEIS-maintained GitHub repository on 9 November 2026 exposes the same prefix; a check on 10 November returned identical entry text. Direct retrieval of the corresponding live b-files and detailed editorial history was unavailable in the author's retrieval environment. The contribution asserted here is a local validated extension of that stated baseline, without a claim of worldwide priority or an assertion that OEIS has incorporated these terms.

The closely related sequence A358862 records the least *value*, rather than its index [2]. Its entry credits Gutkovskiy, 3 December 2022, and Daniel Suteu for terms 12–15 and two PARI programs dated 4 December 2022. Suteu's constructive program enumerates integers with a specified number of distinct prime factors by increasing prime bases and positive powers, then tests polygonality. Its driver also searches successive adjacent value windows. This constructive enumeration and windowing are the central algorithmic prior art used here. The same entry contains Harvey P. Dale's Mathematica program of 9 September 2023 for the first eight terms. The present work is a fresh C implementation of that method, with exact tail bounds, streaming minima and retained coverage evidence; it does not claim to have originated constructive prime-power enumeration.

The distinctive feature of the last minimum is its mixed proof: the computationally completed subtrees are combined with an elementary valuation obstruction for the unfinished part. The obstruction was applied after the expensive search; it was not an early subtree filter of the producer. We first describe the finite search, then prove this obstruction and state the computational result with its precise evidence boundary.

2 A finite and nonredundant search

2.1 Bounds and exact inversion

Let q_i be the ith prime and put $Q_r = \prod_{i=1}^{r} q_i$, with $Q_0 = 1$. If $\omega(v) = n$, then $v \geq Q_n$. Moreover every prime divisor p of such a $v \leq B$ satisfies

$$p \leq \lfloor B/Q_{n-1} \rfloor , \quad (4)$$

because the other $n-1$ distinct primes have product at least Q_{n-1}. The bound remains valid when some exponents exceed one. For an index cutoff K, it is therefore sufficient to construct the n-support integers in $[Q_n, P_n(K)]$.

For fixed n, direct subtraction gives

$$P_n(k+1) - P_n(k) = (n-2)k + 1 > 0. \quad (5)$$

A positive integer v is n-gonal exactly when

$$D = (n-4)^2 + 8(n-2)v = s^2, \qquad k = \frac{s+n-4}{2(n-2)} \in \mathbb{Z}_{\geq 1}, \quad (6)$$

where s is the nonnegative integer square root. This follows by completing the square in (1); the same inverse criterion is given in [4]. A square discriminant alone is insufficient: the divisibility

and positive-index tests are required. The implementation also substitutes the recovered k into (1) before accepting it. For $n = 16$, the equivalent reduced formulas

$$P_{16}(k) = k(7k - 6), \qquad t^2 = 7v + 9, \qquad k = (t + 3)/7 \tag{7}$$

reduce the size of the radicand. Here $t = s/4$ when s is the nonnegative root in (6) for $n = 16$.

2.2 Increasing prime powers and tail pruning

Every integer with n distinct prime divisors has a unique representation

$$v = p_1^{e_1} \cdots p_n^{e_n}, \qquad p_1 < \cdots < p_n, \qquad e_i \geq 1. \tag{8}$$

At a recursive node, m is the current product and r prime bases remain to be selected. For each next prime p in increasing order, the generator tries its powers with exponent $1, 2, \ldots$, and recurses using only primes larger than p. If $t_1 < \cdots < t_r$ are the least allowed remaining primes, then every completion is at least

$$m \prod_{i=1}^{r} t_i. \tag{9}$$

A branch is discarded when this lower bound exceeds B. The bound is monotone as the next prime or its exponent increases, justifying termination of the corresponding loop. The prime list is obtained by a sieve through the inclusive cap (4). Products below a window's lower bound may be constructed, but are not tested for membership in that window.

Proposition 1 (Completeness of the finite generator)**.** *For a finite positive bound B, the preceding procedure generates each integer $v \leq B$ with $\omega(v) = n$ exactly once, before the window lower bound and polygonality tests.*

Proof. The prime cap contains every prime of every admissible tuple by (4). Fix one such tuple. At any of its prefixes, the least-tail product in (9) cannot exceed the actual completion, so its path is never pruned. Monotonicity of the tail bound ensures that every earlier prime or exponent choice also passes its relevant bound, so no preceding loop break prevents reaching the prescribed choice at any depth. Induction on the remaining depth reaches its leaf. Conversely, every leaf has exactly n increasing prime bases with positive exponents, so has support size n. Unique factorization makes two distinct tuples unequal as integers. There are finitely many primes under the cap; each exponent is bounded by $p^e \leq B$; recursion has depth n. Thus all loops terminate. The empty search case $B < Q_n$ follows from the primorial bound and requires no recursive leaves. □

Separate value windows have inclusive lower bound one greater than the previous upper bound. For a fixed n, their coverage therefore joins without a gap or overlap. Since tuple order is not numerical product order, the producer retains a minimum over the whole completed window rather than taking the first polygonal leaf. Equalities (5) and (6) make least value and least index equivalent.

2.3 Arithmetic and persistent minima

The source uses C17 with GCC's `unsigned __int128` extension [7] for products, discriminants and wide counters; the implementation stores prime bases as 32-bit unsigned integers. Before multiplying by p, a bound of the form $m \leq B/p$ prevents overflow. The numerical admission checks bound both the sieve cap by $2^{32} - 1$ and the generic discriminant by $2^{128} - 1$. The implemented input domain

$13 \le n \le 24$ is a numerical capacity statement, not a computed range or a guarantee of feasible running time.

Necessary square-residue tests modulo 64 and 63 precede the exact root calculation at a leaf. The masks are constructed from all square residues, so rejecting a nonsquare residue cannot discard a polygonal value. A floating-point root estimate affects speed only: exact integer-square comparisons correct it in both directions and the acceptance test is integer arithmetic. In particular, no probable-prime test is used to decide membership of an enumerated leaf; its prime support is already explicit.

The tree is partitioned by fixed initial prime-power tuples. The final $n = 19$ jobs use three-base prefixes. OpenMP workers process these disjoint units [7]; a unit may be interrupted and resumed from its last fully processed tuple. In the later producer the cursor order is the lexical order of prime/exponent choices used by the recursion, rather than the order of products. An improved global minimum is durably recorded before a cursor or completion record advances. An interruption before that advance may repeat work, but cannot silently skip an unrecorded smaller witness. A completion record certifies exhaustion of a unit's prescribed traversal; a partial record does not.

The earlier producer, denoted E in Table 1, executed all six $n = 16, 17, 18$ windows and the first three $n = 19$ windows. It used the same increasing-prime-power generation, depth-three partition, tail bounds, leaf masks and exact membership criterion. A completed unit's minimum and certificate were persisted together in its completion record; partial-unit work was discarded and that unit restarted on the next call. The second and third $n = 19$ windows required such continuation. The later producer L retains the tuple cursors and improving minimum records described above. Its first $n = 19$ window has initial index cutoff 81587135064138 and starts at value $P_{19}(81587135064138)+1$, exactly after E's final covered value. Thus E accounts for nine rows and L for three rows of the table. Both producers are retained with their own native verifiers; a separate artifact auditor checks the corresponding histories and their exact join. Integrity of these histories supports provenance, but the mathematical coverage argument also requires the generator, the exact window boundaries, the disjoint unit definitions and their completion or analytic exclusion. The reproducibility material is distributed through the project repository and Zenodo release history [8].

3 Arithmetic exclusion of entire subtrees

Put $L_n(k) = (n-2)k - (n-4)$, which is positive for $n \ge 3$, $k \ge 1$. The two linear factors satisfy

$$\gcd(k, L_n(k)) = \gcd(k, n-4). \tag{10}$$

Here $\gcd(k, -1) = 1$ and $\gcd(k, 0) = k$, so the identity includes $n = 3, 4$. It follows by subtracting $(n-2)k$ inside the gcd. The shared prime support is thus confined to the primes of $n-4$, except that $n = 4$ is the square case. For example (7) gives

$$\omega(P_{16}(k)) = \omega(k) + \omega(7k-6) - \mathbf{1}_{2|k} - \mathbf{1}_{3|k}, \tag{11}$$

because $\gcd(k, 7k-6) = \gcd(k, 6)$. The symbol $\mathbf{1}_E$ is one when E holds and zero otherwise.

For a prime p and a positive integer x, write $v_p(x)$ for the exponent of p in x.

Lemma 2 (Odd-prime obstruction)**.** *Let $n \ge 3$ and let p be an odd prime dividing $n-4$. For every $k \ge 1$,*

$$p \mid P_n(k) \iff p \mid k, \qquad p \mid P_n(k) \implies p^2 \mid P_n(k). \tag{12}$$

In particular $v_p(P_n(k)) \ne 1$.

Proof. Since $n-2 \equiv 2 \pmod{p}$ and 2 is invertible modulo p, $P_n(k) \equiv k^2 \pmod{p}$. Thus $p \mid P_n(k)$ if and only if $p \mid k$. In that case both k and $L_n(k)$ are divisible by p, and division of their product by 2 does not remove a factor of p. This proves the claim. □

The restriction to odd primes is essential to this proof. For example, $P_{10}(2) = 10$ has 2-adic valuation one although $2 \mid 10-4$. For $n=3$ there is no prime divisor of $n-4=-1$. For $n=4$ every odd prime divides $n-4=0$ and the conclusion agrees with $P_4(k)=k^2$; in that case all prime valuations, including the 2-adic valuation, are even. These are elementary consequences of the defining polynomial; no novelty claim is made for the gcd identity or the lemma.

3.1 The two relevant primes for $n = 19$

Here

$$P_{19}(k) = k(17k-15)/2, \qquad n-4 = 15. \tag{13}$$

Lemma 2 applies to $p = 3, 5$. As finite arithmetic checks, the residue lists for $k = 0, \ldots, 8$ modulo 9, and for $k = 0, \ldots, 24$ modulo 25, are respectively

$$(0,1,1,0,7,4,0,4,7), \tag{14}$$

$$(0,1,19,4,6,0,11,14,9,21,0,21,9, \tag{15}$$
$$14,11,0,6,4,19,1,0,16,24,24,16).$$

The second display is one list continued on its next line. Neither $3, 6$ modulo 9 nor $5, 10, 15, 20$ modulo 25 occurs. These lists apply to every integer k by the exact identities

$$P_{19}(k+9) - P_{19}(k) = 9(17k+69), \tag{16}$$

$$P_{19}(k+25) - P_{19}(k) = 25(17k+205). \tag{17}$$

The direct divisibility proof establishes the universal assertion without relying on numerical sampling. Python and PARI/GP also check the displayed lists and period identities independently.

If a fixed prefix contains 3^1 or 5^1, every descendant has that exact valuation: later prime bases are strictly greater than those already fixed. Its entire subtree is therefore empty of 19-gonal values, regardless of how much of the subtree the native program processed. This argument depends on the prime being fixed in the prefix; it does not reject a branch simply because that prime might occur later.

Unit IDs are local to each window: zero-based numbering follows the lexicographic order of $(p_1, e_1, p_2, e_2, p_3, e_3)$ in its prefix map. Changing the upper bound can insert prefixes and change later IDs. In the decisive window the sole unfinished unit, numbered 0, fixes $(2^1, 3^1, 5^1)$. In the final supplement the unfinished units 0 and 110 fix $(2^1, 3^1, 5^1)$ and $(2^2, 3^1, 5^1)$, respectively. All three subtrees are excluded by the lemma. The exponent mapping is checked against the actual prefix definitions and stored cursors, not inferred from a unit's number. The $p = 3$ case alone suffices to exclude all three pending subtrees; $p = 5$ gives an additional sufficient obstruction for them. Across the full prefix maps, the union of the two obstruction classes contains 276 of 577 and 279 of 590 units, respectively. Most of these units were already completed computationally: only the pending units are added as analytic coverage in the proof, so completed units are never counted twice.

The $p = 3$ exclusion was adopted after the search had produced the $n = 19$ witness and had left one native unit unfinished. The $p = 5$ instance was then documented before the final supplement. The unchanged producer still applied its residue masks at leaves; it did not implement these analytic prefix exclusions. Mathematical closure and native completion therefore remain distinct: the last two jobs are retained as incomplete and unsealed.

4 Coverage, certificates and results

4.1 Finite coverage and computational theorem

Table 1 gives the inclusive index cutoffs and prefix counts. For each n, the first value window starts at Q_n, and each later window starts at the previous $P_n(K) + 1$. For $n = 19$ this includes the transition between the earlier and later producers. No index k with $P_n(k) < Q_n$ can satisfy the predicate, since the primorial threshold bounds polygonal values.

Producer	n	K	Closed	Total	Minimum / analytic addition
E	16	68231913854	158	158	none
E	16	152571197612	320	320	119535945060
E	17	506328003064	154	154	none
E	17	876985826602	251	251	none
E	17	1518984009193	393	393	1113316946280
E	18	9762008494901	337	337	4278672701330
E	19	67989279220115	297	297	none
E	19	77519705336978	332	332	none
E	19	81587135064138	343	343	none
L	19	117760885979226	451	451	none
L	19	166539042068893	576	577	153290693815740; unit 0 empty
L	19	170000000000000	588	590	none; units $0, 110$ empty

Table 1: Retained coverage. E and L denote the earlier and later producer. Closed counts distinct natively completed prefix units within each window. The final two rows have full mathematical coverage only after adding the stated analytic exclusions; their native jobs remain incomplete.

Theorem 3 (Validated computational minima)**.** *Subject to the correct execution of the validated prime-power producers and the recorded disjoint coverage in Table 1, together with the exact prefix exclusions of Lemma 2, the minima in* (2) *for* $n = 16, 17, 18, 19$ *and their polygonal values are those in Table 2. For* $n = 19$*, no* k *in the separate interval*

$$166539042068893 < k \leq 170000000000000 \tag{18}$$

satisfies $\omega(P_{19}(k)) = 19$*.*

Computational proof with arithmetic exclusions. Under the execution and coverage hypotheses of the theorem, for $n = 16$ all $158 + 320 = 478$ units are complete and their minimum is the reported index. For $n = 17$ the corresponding counts are $154 + 251 + 393 = 798$, and for $n = 18$ all 337 units are complete. The primorial bound, the adjacent value windows and Proposition 1 account for every possible smaller value. Independent positive certificates establish membership at the reported indices.

For $n = 19$, the first four rows for that n have no hits and complete native coverage through $K = 117760885979226$. The next window has 576 natively complete units out of 577, and the durably recorded global minimum over the union of their processed tuples is 153290693815740. Its positive certificate belongs to complete unit 32, whose three-base prefix is $(2^1, 3^3, 5^2)$. The only remaining unit is 0, excluded by Lemma 2. Thus no smaller matching value remains outside the proved coverage. Strict monotonicity (5) establishes the least index.

The supplement starts at value $P_{19}(166539042068893)+1$ and has 588 complete native units out of 590, with no positive minimum. Its only pending units 0 and 110 are also excluded by the lemma.

This proves (18) empty of matching indices. The native records remain 576/577 and 588/590; no completion record is supplied in place of the arithmetic proof. The negative computational part is not a second, independently implemented exhaustive enumeration. □

n	$a(n)$	$P_n(a(n))$
16	119535945060	100021895128994153154840
17	1113316946280	92960596715494464836371 80
18	4278672701330	14645632068082252436124 1890
19	153290693815740	19973331288934192090890623 6550

Table 2: The four locally admitted terms and the exact same-index polygonal values. The last column is derived from the middle column, rather than from a separate enumeration.

The supplement records an additional finite no-hit interval after the minimum was established; it is not needed to prove that minimum. The producer records minima, rather than all matching indices. In particular, the proof of the least $n = 19$ index does not enumerate every matching index between that minimum and the supplement's lower cutoff. Hence the empty supplement must not be enlarged to the interval $(a(19), 170000000000000]$ without additional evidence. The cutoff concerns P_{19}, and gives no search coverage for P_{20}.

4.2 Independent positive certificates and their limits

For each reported term, an independent Python checker factors the two linear factors by elementary trial division, combines their exponents and accounts for division by 2 in (1). For even orders it can instead factor the already simplified product, as in (7). It compares the prime support and full product with the claimed value, and checks the exact inverse discriminant. Changed-index, changed-value and changed-factor controls are rejected. Separate PARI/GP inputs factor the values and verify primality, support size and inversion. The relevant meanings of `omega`, `bigomega`, `factor`, `isprime` and `ispolygonal` are documented in the official manual [6]. Explicit primality verification avoids treating an unproved large factor from a general factorization routine as automatically prime.

For $k = 153290693815740$, the retained certificate is

$$\begin{aligned} P_{19}(k) = 2 \cdot 3^3 \cdot 5^2 \cdot 7 \cdot 11 \cdot 13 \cdot 17^2 \cdot 19 \cdot 23 \cdot 29 \cdot 31 \cdot 37 \\ \cdot 41 \cdot 43 \cdot 67 \cdot 71 \cdot 73 \cdot 101 \cdot 569. \end{aligned} \tag{19}$$

There are 19 distinct bases. In particular the valuations at 3 and 5 are 3 and 2, consistent with the arithmetic obstruction. The Python check also factors the two linear factors independently of the producer's prime-power traversal. PARI/GP provides another arithmetic backend. Both certify this positive witness, not the assertion that every smaller index fails the predicate.

A retained Python test harness constructs a diagnostic C fixture from the producer and compares its complete small output sets for orders $n = 4, 5, 6, 7$ against a direct distinct-factor-count sieve, with filters both enabled and disabled. These diagnostic orders lie outside the production input domain. This is distinct from the built-in self-test. Further retained tests cover resumed tails, malformed records and interruption/recovery. The small-set oracle counts prime divisors by marking multiples, rather than by the producer's prime-power recursion. These finite tests do not independently enumerate the full production ranges. Replaying Suteu's published GP method on the 13 source terms checks its implementation and the value/index convention; it is reproduction of

the same constructive method, not independent mathematical evidence for all new non-hits. Similarly, an artifact auditor checks integrity, counters, prefix identities, positive products and coverage joins; it does not itself regenerate every omitted leaf.

The resulting certification boundary has three components: mathematical completeness of the finite generator, correct execution and coverage of its computationally closed subtrees, and independently checked arithmetic of the positive witnesses and excluded prefixes. The conditional execution basis in Theorem 3 makes the residual common-mode risk explicit.

5 Consequences for related sequences

Write $b(n)$ for A358862. Since (5) is strictly positive,

$$b(n) = P_n(a(n)) \tag{20}$$

whenever the minimum exists. Conversely, the exact inverse (6) recovers $a(n)$ from $b(n)$. All 13 source pairs for $n = 3, \ldots, 15$ satisfy this relation; for example, $P_3(11) = 66$ and $P_{15}(6066475415) = 239213805711830629680$. Thus the final column of Table 2 is also the locally validated same-index extension of A358862 through 19. It requires no second autonomous search.

A359014, also submitted by Gutkovskiy on 12 December 2022, instead minimizes the index under $\Omega(P_n(k)) = n$; its entry includes Suteu's PARI program and credits his terms 29–34 [3]. This distinction changes the minima already at $n = 3$: $P_3(7) = 28 = 2^2 \cdot 7$ has $\Omega = 3$ and $\omega = 2$. Its longer source table therefore cannot extend the present sequence. More generally, varying n changes both the polynomial and the target support, so none of the relations here converts the finite $n = 19$ coverage into a determination of $a(20)$. No assertion of existence for every n is implied by the finite results.

6 Implementation context and reproducibility

The computation was run on an AMD Ryzen 9 7940HS workstation with eight physical cores and sixteen hardware threads, under Linux Mint 22.3 (x86-64; kernel 6.17.0-42-generic). The production build used GCC 13.3.0, C17, OpenMP and the flags

```
-O3 -march=native -mtune=native -flto -funroll-loops -fomit-frame-pointer
-std=c17 -fopenmp
```

with the retained warning options and the mathematics library. These flags and the original build specification are retained; `-march=native` means that the executable itself is specific to the build host. The final jobs were configured with eight worker threads. This is a configuration, not an eightfold speedup claim. The checkers use Python 3.12.3 standard-library integer arithmetic and PARI/GP 2.15.4. Neither Sidef nor a symbolic algebra package is a dependency of the production enumeration.

As one scoped measurement, the final interval (18) consumed about 9900 seconds of accumulated native search wall time in three calls, including work on prefixes covered analytically but not skipped by the unchanged producer. This is not CPU time or a conservative resource reservation; it also excludes elapsed time between calls. Its stored leaf count is 2791504259667, and its stored exact-root-call count is 52068962797. The lower value cutoff skips membership tests but does not eliminate construction of below-cutoff tuples, so these counters are not counts of distinct admissible indices in the supplement. They must not be used to extrapolate a guaranteed runtime for another order or horizon. No single whole-project timing or peak-memory benchmark is asserted here.

The local reproducibility material retains the source and build specification, the earlier producer needed for its own histories, the 17-term table, exact window and prefix records, positive Python/PARI certificates, modular certificates and validation programs with their inputs and execution outputs. A reconstruction of the proof checks, in order, the historical prefix and value/index map, the producer identities appropriate to each history, the coverage joins and completed units, the analytically excluded prefixes, and the positive factorizations. These steps check the retained computation; a full independent reproduction of the negative search is a stronger and separate task. The repository and archive reference [8] provides the source, data and retained evidence. Its Zenodo series DOI identifies the release history; individual deposits identify fixed versions.

7 Conclusion

The locally validated prefix of A358321 now contains 17 terms, indexed 3 through 19, with four terms beyond the source baseline. The corresponding extension of A358862 is an exact consequence of monotonicity. The last minimum illustrates how an elementary valuation obstruction can complete a finite computational proof by excluding entire unfinished prime-power subtrees. Its negative part still relies on the recorded producer coverage of the other subtrees. The empty final interval is a separate finite statement for order 19; the next undetermined term in this work is $a(20)$.

Data availability

The source, build specifications, complete term table, coverage records, positive certificates, arithmetic exclusions, validation programs, original inputs and execution outputs are available through the GitHub repository and Zenodo release history cited in References. This paper is supplied as a monolithic TeX source and matching PDF with the repository package. The Zenodo citation uses the series (concept) DOI; individual deposits in its version history identify fixed releases.

AI use disclosure

Large language models were used as research and drafting assistants during the A358321 workflow, including OpenAI Codex and external LLM-based review systems presented as ChatGPT, Claude, DeepSeek, Gemini, Grok, Kimi, Qwen, MiniMax and GLM/Z.ai. These are recorded service or family attributions; exact backend identities and model versions were not consistently authenticated. One manuscript review was obtained through a Mistral-hosted service and identified its model family as GLM; that backend attribution was not independently authenticated. Meta AI was also used as a service in preliminary analysis; its underlying model family is not assumed. The assistance and review reports are retained privately.

Mathematical analysis. AI assistance was used to examine the distinct-factor predicate, source relationships, finite enumeration bounds and arithmetic subtree exclusions. The accepted statements are supported by the defining identities, preserved OEIS source records, explicit proofs, result tables and Python/PARI arithmetic certificates. Repeated source values in an AI response do not constitute independent confirmation.

Algorithm and code. Codex assisted implementation and validation design. Reports identifying Claude and MiniMax contributed to code review. Final computational claims are tied to the retained

source/build identities, recorded coverage, small exact-set comparisons, cursor/recovery tests and independent positive certificates. These checks do not turn LLM review or positive arithmetic certification into a second exhaustive negative enumeration. The arithmetic prefix exclusions were applied after the search and are not attributed to an earlier implementation feature.

Manuscript. Codex assisted drafting, literature verification, arithmetic cross-checks and internal examination. External LLM manuscript reviews of the initial draft and focused reviews of the revised draft were assessed against the canonical term table, coverage records, original source exports, certificate programs and execution outputs. Their reported arithmetic checks and static prefix-map replays are distinct from an independent exhaustive replay of the negative search. Final numerical claims rely on those local artifacts and validation, not on conversational assertions. These reports do not establish independent reproduction of the full negative enumeration or acceptance by a publication venue.

The author, Carlo Corti, takes full scientific responsibility for the mathematical statements, algorithm, implementation, numerical results, validation boundaries and conclusions of this paper.

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